

ENGINEERING REBOOT 2026

Week 02 – Day 09 Field Manual v3.3

Functions and Code Reuse

PHASE 0 – MISSION BRIEF

Objective

Learn how to organize programs into reusable functions.

Functions allow software engineers to reduce duplication, improve maintainability, and create modular software components.

Core Concepts

- def
- parameters
- return values
- function calls
- code reuse

Engineering Applications

- Software libraries
- Automation frameworks
- Embedded firmware
- API wrappers
- Business logic modules

Learning Objectives

- How to define a function
- How to call a function
- How parameters work
- How return values work
- Why code reuse matters
- Basic program modularization

Deliverable Inventory

1. greeting_function.py
2. calculator_function.py
3. temperature_converter.py
4. Notes.md
5. Journal.md
6. Evidence Package

PHASE 1 – GREETING FUNCTION

Filename

greeting_function.py

Implement

- Create function greet()
- Function displays a greeting message
- Call the function from the main program

Validation Matrix

Expected Output:

Hello from my first function!

Evidence Checkpoint

- greeting_function-Code.png
- greeting_function-Output.png

- Validation complete
- Evidence captured

PHASE 2 – CALCULATOR FUNCTION**Filename**

calculator_function.py

Implement

- Create function add_numbers()
- Accept two parameters
- Return the calculated result
- Display returned value

Validation Matrix

Input: 10, 15

Expected Output: $10 + 15 = 25$

Evidence Checkpoint

- calculator_function-Code.png
- calculator_function-Output.png

- Validation complete
- Evidence captured

PHASE 3 – TEMPERATURE CONVERTER**Filename**

temperature_converter.py

Implement

- Create function celsius_to_fahrenheit()
- Accept Celsius value as parameter
- Return Fahrenheit value
- Display result

Validation Matrix

0 → 32.0

100 → 212.0

Evidence Checkpoint

- temperature_converter-Code.png
- temperature_converter-Output.png

- Validation complete
- Evidence captured

PHASE 4 – NOTES

Create: Week-02/Notes/Day-09/Notes.md

Required Sections:

- Concepts Learned

- Key Commands
- Key Observations
- Lessons Learned
- Questions For Future Study

Suggested Topics:

- What is a function?
- Why use functions?
- What is a parameter?
- What is a return value?
- How does code reuse improve software?

■ Notes completed

PHASE 5 – JOURNAL

Create: Week-02/Journal/Day-09/Journal.md

Required Sections:

- Summary
- Challenges Encountered
- Solutions Applied
- Confidence Assessment
- Reflection
- Tomorrow Preparation

Suggested Reflection:

Which concept required more thought: Parameters or Return Values?

■ Journal completed

PHASE 6 – GIT OPERATIONS

```
git status
git add .
git commit -m "Complete Day 09 function exercises"
git push
git fetch
git status
git branch -vv
```

Preferred End State

Your branch is up to date with 'origin/main'
nothing to commit, working tree clean

Evidence Checkpoint

Git-Operations.png

Must Show:

- git add .
- git commit
- git push
- git fetch
- git status
- git branch -vv

■ Git operations completed

- Repository synchronized
- Evidence captured

END-OF-DAY SUCCESS CRITERIA

- greeting_function.py completed
- calculator_function.py completed
- temperature_converter.py completed
- All validation completed
- Exercise evidence captured
- Notes completed
- Journal completed
- Git workflow completed
- Git-Operations.png captured
- Remote synchronization verified
- Working tree clean

MISSION COMPLETE ✓

DAY 10 PREVIEW

Topic: Mini Project Integration

Expected Deliverables:

- calculator_application.py
- project_notes.md

Core Concepts:

- functions
- loops
- conditionals
- user input
- program structure
- integration of concepts